IMPERIAL COLLEGE OF SCIENCE, TECHNOLOGY AND MEDICINE

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

EXAMINATIONS 2024

MATHEMATICS FOR SIGNALS AND SYSTEMS SOLUTIONS

SOLUTIONS

1. (a) i. Vectors in $\mathbb{R}^4$ with $x_1 + x_2 + x_3 = 0$ and $x_3 + x_4 = 0$ have the following form:

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ -x_1 - x_2 \\ x_1 + x_2 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 0 \\ -1 \\ 1 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 1 \\ -1 \\ 1 \end{bmatrix}$$

So a basis is $\mathbf{n}_1 = [1, 0, -1, 1]^T$, $\mathbf{n}_2 = [0, 1, -1, 1]^T$.

ii. Let $\mathbf{A}$ be such matrix. Since $\mathbf{A}\mathbf{n}_i = 0$, $i = 1, 2$, the matrix must have 4 columns but can have an arbitrary number of rows. We pick $\mathbf{A} \in \mathbb{R}^{2\times 4}$ and a possible matrix is:

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & 2 & 1 \\ 3 & 3 & 2 & -1 \end{bmatrix}.$$

iii.

$$\mathbf{A} = \begin{bmatrix} \mathbf{n}_1 & \mathbf{n}_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -1 & -1 \\ 1 & 1 \end{bmatrix}.$$

(b) i. We need to find two linearly independent vectors such that $\mathbf{A}\mathbf{n}_i = 0$, $i = 1, 2$. A possible solution is $\mathbf{n}_1 = [0, 1, 1]^T$, $\mathbf{n}_2 = [1, 0, 3]^T$.

ii. We note that $V^\perp$ has dimension 1. So we need to find only one vector, $\mathbf{u}$, orthogonal to $\mathbf{n}_1$ and $\mathbf{n}_2$. Clearly the row of $\mathbf{A}$ is such solution is $\mathbf{u} = [3, 1, -1]$.

iii. By definition $\mathbf{P}_1 = \mathbf{u}(\mathbf{u}^H\mathbf{u})^{-1}\mathbf{u}^H$ with

$$\mathbf{u} = \begin{bmatrix} 3 \\ 1 \\ -1 \end{bmatrix}.$$

So

$$\mathbf{P}_1 = \frac{1}{11} \begin{bmatrix} 9 & 3 & -3 \\ 3 & 1 & -1 \\ -3 & -1 & 1 \end{bmatrix}.$$

iv. The easiest way to find $\mathbf{P}_2$ is by noting that $\mathbf{P}_2 = \mathbf{I} - \mathbf{P}_1$. So

$$\mathbf{P}_2 = \frac{1}{11} \begin{bmatrix} 2 & -3 & 3 \\ -3 & 10 & 1 \\ 3 & 1 & 10 \end{bmatrix}$$

(c) The matrix has rank 1 and is of size 2×2, so all the four subspaces have dimension 1. A possible basis for the null space is $\mathbf{n} = [2, -1]^T$. A possible basis for the range space of $\mathbf{A}$ is one column of $\mathbf{A}$ so $\text{range}(\mathbf{A}) = \text{span}([1, 3]^T)$. A possible basis for the range space of $\mathbf{A}^T$ is one row of $\mathbf{A}$ so $\text{range}(\mathbf{A}^T) = \text{span}([1, 2]^T)$ and finally $\text{null}(\mathbf{A}^T) = \text{span}([3, -1]^T)$.

2. (a) We are effectively asked to find the null space of $\mathbf{A}$ and we know that this is a subspace of dimension $d = n - rank(\mathbf{A})$ where n is the number of columns of $\mathbf{A}$. In this case $n = 12$ so dimension of the null space is $d = 4$. So 4 linearly independent solutions to the homogeneous system can be found.

   (b) The rank of the matrix is the number of columns of $\mathbf{A}$ minus the dimension of the null space. So in this case the rank is 8 which corresponds to the number of rows. So the matrix is full row rank and therefore a solution to $\mathbf{Ax} = \mathbf{b}$ always exists (and is not unique). Specifically the range space of $\mathbf{A}$ span $\mathbb{R}^6$ and $\mathbf{b} \in \mathbb{R}^6$ so a solution always exists.

   (c) i. The coefficients $\mathbf{c} = [\alpha, \beta]^T$ that give us the least square fit are the solution of:

$$\mathbf{A}^T\mathbf{Ac} = \mathbf{A}^T\mathbf{b},$$

where

$$\mathbf{A} = \begin{bmatrix} x_1 & 1 \\ x_2 & 1 \\ x_3 & 1 \end{bmatrix}$$

and

$$\mathbf{b} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}.$$

In this case

$$\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 2 & 1 \\ 3 & 1 \end{bmatrix}$$

and

$$\mathbf{b} = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}.$$

Therefore, the coefficients are the solution of

$$\begin{bmatrix} 14 & 6 \\ 6 & 3 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} 14 \\ 6 \end{bmatrix}$$

So $\alpha = 0.5$, $\beta = 1$.

   ii. In this case, if we write the equation in the right order we realize that the problem is completely symmetric, with

$$\mathbf{B} = \begin{bmatrix} y_1 & 1 \\ y_3 & 1 \\ y_2 & 1 \end{bmatrix}$$

and

$$\mathbf{b_2} = \begin{bmatrix} x_1 \\ x_3 \\ x_2 \end{bmatrix}.$$

Therefore $\gamma = 0.5$, $\delta = 1$.

(d) i. The least-squares solution is found by solving the normal equation:

$$\mathbf{A}^T\mathbf{A}\mathbf{x} = \mathbf{A}^T\mathbf{b}.$$

In this case the solution is not going to be unique given that $\mathbf{A}$ is not full column rank. We have:

$$\begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$$

and all possible least-squares solutions are of the form $\mathbf{x}_{LS} = [x_1, 1.5 - x_1]^T$ with $x_1 \in \mathbb{R}$.

ii. We can compute the pseudo-inverse of $\mathbf{A}$ to find the least-squares minimum norm solution. However, in this case, we can directly minimize $\|\mathbf{x}_{LS}\|^2 = 2x_1^2 + 2.25 - 3x_1$. The minimum is achieved when $x_1 = 3/4$. Consequently, $\mathbf{x}_{LSMN} = [3/4, 3/4]^T$.

3. (a) i. The matrix has rank one. We have that:

$$\mathbf{A} = \mathbf{x}\mathbf{x}^T = \begin{bmatrix} x_1^2 & x_1x_2 & \cdots & x_1x_n \\ x_2x_1 & x_2^2 & \cdots & x_2x_n \\ \vdots & \vdots & \ddots & \vdots \\ x_nx_1 & x_nx_2 & \cdots & x_n^2 \end{bmatrix}$$

The matrix $\mathbf{A}$ is the outer product of $\mathbf{x}$ with itself. So every row (or column) of $\mathbf{A}$ is a scalar multiple of the vector $\mathbf{x}$. This implies that all rows of the matrix are linearly dependent because each row is just a scaled version of $\mathbf{x}$. Thus, there is only one linearly independent row in $\mathbf{A}$.

ii. A. Here $\mathbf{A}$ is the sum of two rank one matrices so its rank is at most 2. If $\mathbf{y}$ is a rescaled version of $\mathbf{x}$ then the rank is 1.

B.

$$\mathbf{A} = \mathbf{x}\mathbf{x}^T + \mathbf{y}\mathbf{y}^T = \begin{bmatrix} 1.25 & 0.5 \\ 0.5 & 2 \end{bmatrix}$$

The eigenvalues are $\lambda_1 = 2.25$ and $\lambda_2 = 1$. The matrix is symmetric, so the two eigenvectors are orthogonal. Given that $\mathbf{u}_2 = [1, -0.5]^T$ then $\mathbf{u}_1 = [0.5, 1]^T$.

(b) i. We have

$$\mathbf{A}^T\mathbf{A} = \begin{bmatrix} 25/2 & 25/2 \\ 25/2 & 25/2 \end{bmatrix}.$$

The matrix is rank one so, $\lambda_2 = 0$, moreover $\lambda_1 + \lambda_2 = \text{trace}(\mathbf{A}) = 25$, therefore, $\lambda_1 = 25$. This means that the only non-zero singular value of $\mathbf{A}$ is $\sigma_1 = 5$.

ii. $\mathbf{v}_1^T = \mathbf{u}_1^T\mathbf{A}/\sigma_1 = [1/\sqrt{2}, 1/\sqrt{2}]^T$. Moreover, $\mathbf{v}_2$ is orthogonal to $\mathbf{v}_1$ and with norm one, so $\mathbf{v}_2 = [1/\sqrt{2}, -1/\sqrt{2}]^T$. With a similar argument we obtain $\mathbf{u}_2 = [1/\sqrt{2}, -1/\sqrt{2}, 0]^T$ and $\mathbf{u}_3 = [0, 0, 1]^T$.

iii. The pseudo inverse is

$$\mathbf{A}^\dagger = \mathbf{V}\mathbf{\Sigma}^\dagger\mathbf{U}^H = \begin{bmatrix} 0.1 & 0.1 & 0 \\ 0.1 & 0.1 & 0 \end{bmatrix}.$$

So

$$\mathbf{x}_{LSMN} = \mathbf{A}^\dagger\mathbf{b} = [0.1, 0.1]^T.$$

(c) The columns of $\mathbf{Q}$ should sum up to one. Therefore

$$\mathbf{Q} = \begin{bmatrix} a & b \\ 1-a & 1-b \end{bmatrix}.$$

Moreover, each entry of $\mathbf{Q}$ represents a probability so $0 \leq a, b \leq 1$. Finally $\mathbf{u}_1 = [1/2, /2]^T$ is the steady state. This means $\mathbf{Q}\mathbf{u}_1 = \mathbf{u}_1$. So a consequence of that is that $a + b = 1$. Besides $a = b = 0.5$, any solution satisfying these constraints is valid. We need to avoid $a = b = 0.5$ because it would lead to a rank deficient matrix and we are told $\mathbf{Q}$ is full rank.